Constraint Programming Optimization

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1st Problem : Frequency Allocation

TODO

2nd Problem : Integrated Healthcare Timetabling Competition (IHTC 2024)

1. Description

This project focuses on modeling and solving a simplified version of the scheduling problem introduced in the IHTC 2024, using Constraint Programming. The objective is to schedule the admission of patients into a hospital over a given planning horizon.

For each admitted patient, the model must determine:

- The admission day.
- The assigned room.
- The assigned operating theater.

By respecting some hard constraints given in the competition.

2. Model

Decision Variables :

```
x_admitted[i]: 1 if patient $i$ is admitted, 0 otherwise.

x_day[i][d]: 1 if patient $i$ is admitted on day $d$.

x_room[i][r]: 1 if patient $i$ is assigned to room $r$.

x_ot[i][o]: 1 if patient $i$ is assigned to operating theater $o$.
```

Sum constraints iterate over the planning to ensure that daily room capacities, OT availabilities and surgeon schedules are not exceeded.
Gender mixing is prevented by enforcing that the sum of males or the sum of females in a room on any given day is strictly zero.

Objective function :

```
unplanned_optionals = [
    (1 - x_admitted[i]) for i, p in enumerate(patients_data) if not p.get("mandatory", False)
]
minimize(Sum(unplanned_optionals))
```

This corresponds to the S8 objective, which aims to minimize the number of unscheduled non-mandatory patients

3. Problems Encountered

Handling JSON Parsing Errors: We couldn't use PyCSP3's standard -data= because it crashes on the asymmetric IHTC JSON files (e.g. missing keys like incompatible_room_ids cause a NamedTuple TypeError). To get around this limitation, we handled the JSON parsing manually and passed the file path through our own -mydata= argument instead

4. Optimization

Pre-filtering To handle massive instances we introduced a pre-filtering step. Before generating constraints, we analyze the input data to identify days where a patient's surgeon is unavailable, when all operating theaters lack capacity for the surgery. For these impossible days, $x_day[i][d]$ is set to 0.

Results

Instance	Resolution Time (s)	First Sol. Time (Bound)	No. of Solutions	Best Objective (S8)
i29	194	119 (208)	6	203
i28	407	301 (191)	2	190
i25	406	29 (211)	41	101
i20	506	no solution found		
i15	207	14 (102)	36	67
i10	207	21 (107)	44	64
test01	26	1.47 (31)	18	8
test09	60	17.2 (58)	7	52

Result Analysis

Overall, the solver does a pretty good job on most instances, even though it clearly hits its limits on the hardest problems.

Usage example

```
python -X utf8 .\projet.py -mydata=\test\test09.json -solve -s=all
```